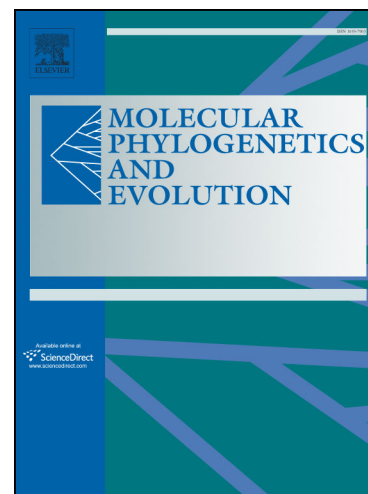


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Diversification and biogeographic history of African dormice (genus *Graphiurus*) revealed by ultraconserved elements and mitochondrial data

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Abstract

The sub-Saharan Africa endemic dormouse genus *Graphiurus* is a morphologically diverse group of rodents that has long lacked a genus-wide genetic framework, hindering inference of their evolution and biogeography. Here, we assembled the first comprehensive, range-wide genomic dataset for *Graphiurus*, including ultraconserved elements and the cytochrome b gene. We used phylogenetic reconstruction, divergence-time estimation, and ancestral area reconstruction to clarify biogeographic histories and evaluate how historical range dynamics may have influenced lineage diversification. *Graphiurus* is the earliest-diverging lineage within Gliridae sister to all other extant genera. Within *Graphiurus*, we recovered 24 molecular operational taxonomic units (MOTUs) forming two reciprocally monophyletic, deeply divergent clades: a West African lineage comprising three MOTUs and a pan-sub-Saharan Africa lineage comprising 21 MOTUs. Genetic distances between these clades are equal to or greater than those used to distinguish genera. Diversification within *Graphiurus* started in the middle Miocene, peaking in a rapid radiation during the Plio-Pleistocene. These diversification events coincided with increased climatic instability that fragmented forests into isolated refugia. Ancestral-area reconstructions suggest a *Graphiurus* origin in the Upper Guinean rainforest, followed by jump dispersal across the Afrotropics, with most subsequent divergences occurring in East Africa. As the first comprehensive phylogenetic analysis of *Graphiurus*, our study underscores the urgent need for an integrated taxonomic revision of the genus that couples genomic data with detailed morphology and critical re-examination of type material to resolve species limits and formally describe the recovered MOTUs. Many MOTUs appear range restricted, underscoring vulnerability to ongoing habitat loss within montane refugia.

Keywords—adaptive radiation, African dormice, *CYTB*, evolutionary history, *Graphiurus*, historical biogeography, phylogenomics, species delimitation, ultraconserved elements.

1 Introduction

African dormice (*Graphiurus*; Graphiurinae: Gliridae) are distributed widely across sub-Saharan Africa (Wilson and Reeder, 2005; Monadjem et al., 2015; Wilson et al., 2017). Described *Graphiurus* species exhibit significant morphological and ecological overlap which has complicated species delimitation and obscured the genus evolutionary processes (Krystufek et al., 2004; Holden and Levine, 2009; Hennekam, 2022). While the monophyly of *Graphiurus* species with respect to other glirids is well established, the timing, mechanisms, and drivers of its diversification and dispersal remain poorly understood (Holden, 1996; Montgelard et al., 2003; Nunome et al., 2007; Holden and Levine, 2009). Fossil records and early phylogenetic studies have suggested a pre-Pliocene colonization of Africa by *Graphiurus* after it diverged from a Eurasian sister clade (deMenocal, 1995; Montgelard et al., 2003). It then colonized sub-Saharan Africa through adaptive radiation and rapid diversification likely from a single ancestral form, but

with minimal morphological changes among the descendants (Genest-Villard, 1978; Holden, 1996; Montgelard et al., 2003). However, assessing this hypothesis is hindered by the lack of a robust phylogenetic and biogeographic framework at the genus level (Lu et al., 2021).

The monophyly of *Graphiurus* is well-supported at the dormouse family level, with the lineage distinctly separated from other genera morphologically and genetically (Vianey-Liaud and Jaeger, 1996; Bentz and Montgelard, 1999; Montgelard et al., 2003). However, these early studies also show short evolutionary branches among species, suggesting a rapid speciation event associated with their colonization of diverse habitats (Montgelard et al., 2003). Overall, *Graphiurus* exemplifies a relict rodent lineage extending from the Oligocene to the Miocene (Hartenberger and Thaler, 1963; Haq et al., 1987; Pickford and Morales, 1994; Mead, 2001; Wilson and Reeder, 2005; Fortelius et al., 2006; Casanovas-Vilar et al., 2011; Lu et al., 2021), unlike the late Cenozoic radiations of most of Africa's rodent diversifications (Mikula et al., 2021; Nicolas et al., 2021; Montgelard et al., 2023). In this respect, *Graphiurus* is a good model group for investigating relatively older biogeographic processes in the Afrotropics, and the warrants modern molecular investigations to disentangle its diversification pattern–process dynamics.

Among glirids, *Graphiurus* is the most species-rich and the only genus endemic to sub-Saharan Africa (Wilson et al., 2017; Mammal Diversity Database, 2025). However, species phylogenetic arrangement within the genus remains unresolved (Holden, 1996; Holden and Levine, 2009; Monadjem et al., 2015; Wilson et al., 2017), with newer classifications consistently rendering the existing nomenclature inconsistent (Fig. 1; Fig. S1), as it is characteristic of the classical systematics of African small mammals (Taylor et al., 2019). For example, the three *Graphiurus* subgenera recognized in Mammal Species of the World Third Edition—*Graphiurus*, *Claviglis*, and *Aethoglis*—were originally distinguished using combinations of morphological and ecological traits (Pavlinov and Potapova, 2003; Wilson and Reeder, 2005). *Aethoglis* (monotypic for *G. nagtglasii*) are distinguished mainly by an elongated skull with narrow nasals projecting anteriorly beyond the incisors, *Claviglis* (monotypic for *G. crassicaudatus*) by a shorter skull with wider braincase and non-projecting nasals, while the nominotypical subgenus *Graphiurus* comprise the remaining species (Pavlinov and Potapova, 2003; Wilson and Reeder, 2005). However, these characters have been applied inconsistently and often vary among putative species, limiting their reliability for diagnosing species boundaries at the genus level. Although cranial and dental morphology remain the most taxonomically informative traits between *Graphiurus* species, they have limited standalone diagnostic stability across the genus, especially when no *a priori* nearest relative is selected for comparisons (Holden and Levine, 2009; Hennekam, 2022). Robust, genus-wide molecular datasets integrated with morphology and geography data are needed to stabilize *Graphiurus* species-level taxonomy.

Despite the wave of recent integrative revisions that have continued to clarify the taxonomy, phylogeny, and biogeography of many African rodents [e.g., Demos et al. (2013); (Bryja et al., 2014a); Bryja et al. (2014c); Demos et al. (2014); Bohoussou et al. (2015); Bryja et al. (2019); Krasova et al. (2019); Giarla et al. (2020); Kerbis Peterhans et al. (2020); Nicolas et al. (2020); Hánová et al. (2021); Nicolas et al. (2021); Onditi et al. (2021); Uhrová et al. (2021); Voelker et al. (2021); Mizerovská et al. (2023); Bryja et al. (2024); Meheretu et al. (2024);

Šumbera et al. (2024)], *Graphiurus* remains an outlier. Species boundaries and evolutionary relationships within the genus are still poorly resolved, as most described exhibit few consistent morphological synapomorphies to anchor a genus-wide classification (Lu et al., 2021). To date, there is no robust molecular phylogeny to validate the classical morphology-based cladograms that continue to underpin supermatrix phylogenies (Kumar et al., 2017; Upham et al., 2019) and taxonomic checklists (Edler et al., 2017; Marsh et al., 2022). There is a need for a comprehensive, multilocus phylogenetic framework to untangle the systematic placement of *Graphiurus* lineages.

Moreover, understanding the phylogenetic relationships, evolutionary patterns, and ecological adaptations of *Graphiurus* is vital not only for resolving taxonomic uncertainties and guiding conservation efforts but also for informing public health and biomedical research. Some *Graphiurus* species are thought to be the primary hosts of the monkeypox virus; the 2003 US monkeypox outbreak was linked to the shipment of small mammals from Ghana (Hutson et al., 2007; Levine et al., 2007), among which *Graphiurus* were included, bringing renewed interest in its biology to better manage human monkeypox outbreaks (Hutson et al., 2007; Schultz et al., 2009; Han et al., 2015; Doty et al., 2017; Hakim and Widyaningsih, 2023; Sun et al., 2023). Limited knowledge of putative vectors' taxonomic status, evolutionary history, and ecological variability hampers control efforts since disease physiology and the success of interventions can vary across different vector lineages and distributional limits (Cuypers et al., 2023). Uncertainties about dormice taxonomy, evolution, and ecology undermine effective disease control, and focused, integrative work is essential for targeted zoonosis-control strategies.

Here, we assembled the most comprehensive genomic dataset for the African dormouse genus *Graphiurus*, including 331 georeferenced specimens sequenced for the mitochondrial cytochrome b (*CYTB*) gene and 56 for genome-wide ultraconserved elements (UCEs). Because of their deep mutational conservation across vertebrate lineages, UCEs generate large, cost-effective data partitions that consistently yield well-resolved phylogenies at various taxonomic scales (Faircloth et al., 2012; Cummins et al., 2024). On the other hand, *CYTB* remains one of the most extensively sampled and phylogenetically informative mitochondrial markers in mammalian phylogenetics, combining an extensive historical sequence archive with phylogenetic signal across multiple taxonomic levels (Baker and Bradley, 2006; Tobe et al., 2010; Caraballo et al., 2021). Integrating these datasets, we applied phylogenomic and divergence dating approaches to provide a detailed analysis of the phylogenetic, biogeographic, and evolutionary dynamics within *Graphiurus* to resolve conflicting signals between limited genetic datasets and morphology-based taxonomic classifications. The main objectives of this study were (1) to reevaluate the phylogeny of African dormice in the genus *Graphiurus* at a pan-African scale using *CYTB* and UCE data; (2) to resolve evolutionary history and biogeographic patterns from a genomic perspective; and (3) to investigate the role of temporal habitat changes in driving the speciation and radiation of this genus.

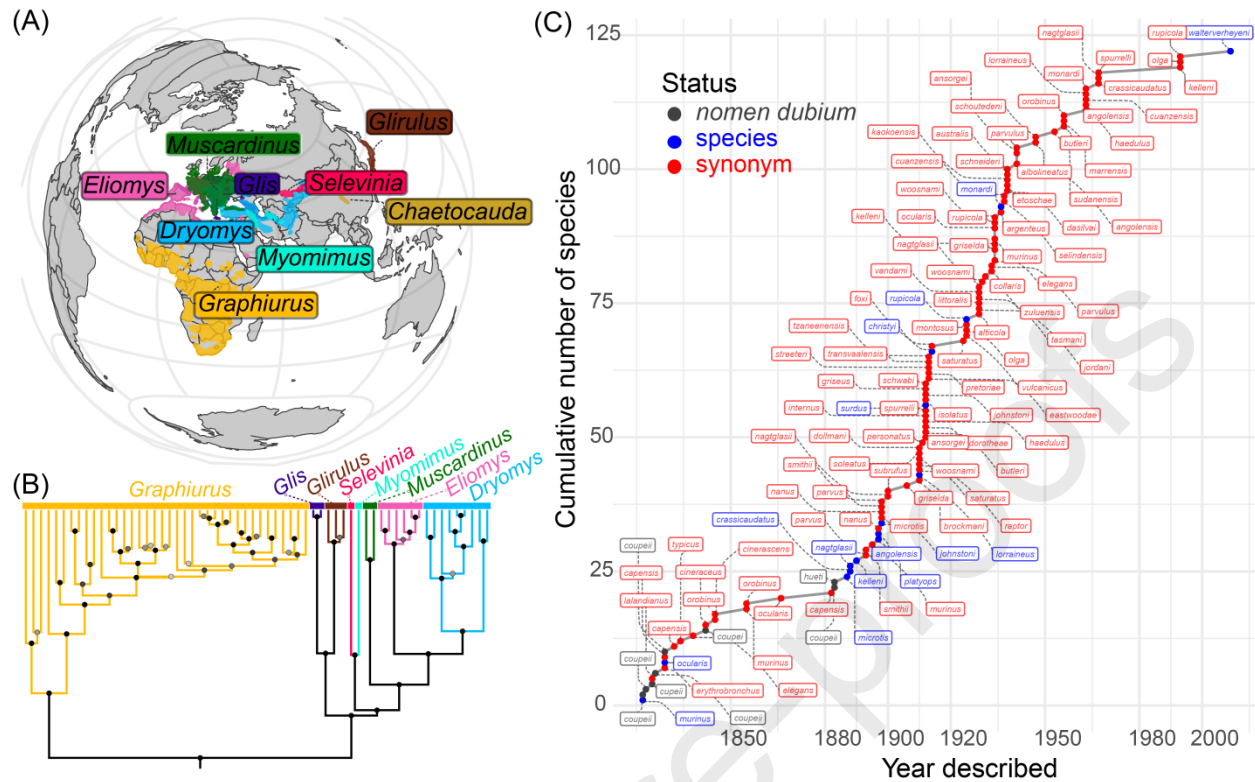


Fig. 1. Overview of taxonomy, distribution, and species relationships in the dormouse family Gliridae. (A) General-level geographic range extents based on IUCN spatial data (<https://www.iucnredlist.org/resources/spatial-data-download>). (B) A simplified phylogeny of Gliridae genera inferred from a Bayesian analysis of cytochrome b sequences compiled for this study; colors correspond to the range map in panel A. (C) Temporal accumulation of *Graphiurus* species names based on the Mammal Diversity Database (v2.1).

2 Methods

2.1 Sampling

The study spans the known distribution range of *Graphiurus* species in sub-Saharan Africa, from West African to Congo Basin forests, East African highland forests and savannas, and southern African biomes (Fig. 2, Fig. S2). We obtained tissue samples through field surveys and museum collections spanning 23 countries (Fig. 2). Tissue samples stored in 95% ethanol or dimethyl sulfoxide were used for DNA extraction. The complete list of samples, collections/collectors, and locality details is summarized in Table S1.

All fieldwork complied with legal regulations in each country, and sampling was conducted with the permission of the responsible national organizations. Animal handling and processing adhered to the permit conditions and the ethical guidelines for using animals in research (Sikes and the Animal Care Use Committee of the American Society of Mammalogists, 2016).

In total we analyzed a total of 348 unique samples: 341 with *CYTB* data and 59 with UCE data. The *CYTB* dataset comprised 331 *Graphiurus* samples, including 74 sequences downloaded from GenBank (Fabre et al., 2013; Nakamura et al., 2013; Bletsa et al., 2018; Petruzela et al., 2019; Krasova et al., 2021; Tomass et al., 2021; Onditi et al., 2022; Martynov et al., 2023). In addition to *Graphiurus*, the *CYTB* dataset included three Glirinae species (one *Glirulus japonicus* and two *Glis glis*), six Leithiinae species (*Dryomys laniger*, *Dryomys nitedula*, *Eliomys quercinus*, *Muscardinus avellanarius*, and *Myomimus roachi*), and one Sciurinae (Sciuridae) species (*Sciurus vulgaris*; NCBI RefSeq NC_002369.1) which was used as an outgroup to the Gliridae samples. Using preliminary *CYTB* phylogenies to guide per-MOTU sampling, we selected 59 samples for UCE sequencing, comprising 56 *Graphiurus* samples and one sample each of *Glis glis*, *Dryomys nitedula*, and *Muscardinus avellanarius*.

2.2 *CYTB* sequencing and alignment processing

Total DNA was extracted from tissue samples with commercial kits (innuPREP DNA Mini Kit, Innuscreen, Germany) following the manufacturer's protocol, and the *CYTB* gene was Sanger sequenced (Bryja et al., 2014c). We amplified the complete *CYTB* gene through PCR with the primers L14723 and H15915 (Lecompte et al., 2002). The PCR amplification was carried out with the multiplex PCR kit (Qiagen, Germany), with an initial denaturation step at 95°C for 15 minutes, denaturation for 35 cycles of 30 seconds at 95°C, annealing for 30 seconds at 50°C, extension for 1 minute at 72°C, and a final extension at 72°C for 10 minutes. We sequenced the full *CYTB* segment for 258 new samples. Because we also aimed to reconstruct the position of *Graphiurus* within Gliridae using all available genetic data, we downloaded GenBank sequences that mentioned “Gliridae” AND “*CYTB*” or “*CYTB*” in the descriptions, which included a total of 492 sequences as of 21 March 2025, 74 of which belonged to *Graphiurus* (Table S2). The combined sequences were aligned using MAFFT v7.525 (Katoh et al., 2002; Katoh and Standley, 2013) and MACSE release 2.07 (Ranwez et al., 2018), with manual inspection in Aliview (Larsson, 2014) to confirmed that there were no premature stop codons. Alignment summary statistics were generated using SEGUL (Handika and Esselstyn, 2024).

The new *CYTB* sequences have been deposited in GenBank under accession numbers PX513364–PX513453.

2.3 UCE sequencing and alignment processing

Genomic DNA was extracted from tissue samples (frozen ethanol- or EDTA-preserved) using an innuPREP DNA Mini Kit (Innuscreen, Germany) following the manufacturer's recommended protocol. The DNA samples were separated on a 1.5% agarose gel, and their concentrations were measured using a Qubit BR DNA kit. All high-quality, high-quantity DNA was shipped to Rapid Genomics (Gainesville, FL, USA) for library preparation and sequencing. The MyBaits UCE Tetrapods 5Kv1 probe set (MYcroarray, Arbor Biosciences, Ann Arbor, MI, USA) was used for target enrichment.

We processed raw reads on the MetaCentrum Grid Computing portal (Czech Republic). We followed the PHYLUCE v1.7.3 workflow (<https://phyluce.readthedocs.io>) (Faircloth, 2013). The raw reads were cleaned with Illuminaprocessor 2.0, a wrapper around Trimmomatic (Del Fabbro et al., 2013; Bolger et al., 2014), to remove adapter contamination and trim low-quality bases.

The cleaned reads were de novo assembled with SPAdes (v4.0.0) (Prjibelski et al., 2020) using default parameters. The resulting contigs were matched to the MyBaits UCE Tetrapods 5Kv1 probe set using `phyluce_assembly_match_contigs_to_probes`, and locus recovery across taxa was summarized with `phyluce_assembly_get_match_counts`. We built a monolithic FASTA per-locus set using `phyluce_assembly_get_fastas_from_match_counts` and split it into single-locus files with `phyluce_assembly_explode_get_fastas_file`. Each locus was aligned with MAFFT through `phyluce_align_seqcap_align` and removed poorly aligned regions with the Gblocks wrapper `phyluce_align_get_gblocks_trimmed_alignments_from_untrimmed`. To standardize headers, we removed locus identifiers from sequence labels using `phyluce_align_remove_locus_name_from_files`. We inspected alignment statistics using `phyluce_align_get_align_summary_data`.

We generated completeness-filtered datasets by excluding loci below specified taxon-occupancy thresholds (50%, 70%, 80%, 90%, 95%) with `phyluce_align_get_only_loci_with_min_taxa`. We then concatenated the curated alignments into a supermatrix, along with accompanying partitions/charset files using `phyluce_align_concatenate_alignments` for downstream phylogenomic analyses.

2.4 Phylogenetic reconstruction

We first thinned the full *CYTB* dataset to eliminate sequence redundancy; specifically, we clustered near-identical sequences based on pairwise similarity and retained the longest sequence as the cluster representative, yielding 110 sequences. For the UCE data, we explored the optimal balance among locus coverage, phylogenetic robustness, and computational demands in downstream analyses using the completeness-filtered datasets. In general, neither the number of loci nor the total alignment length demonstrated a direct correlation with increased phylogenetic informativeness, although more stringent filtering parameters yielded higher completeness and lower rates of missing data, resulting in fewer retained loci and lower bootstrap support values (Fig. S3 and Table S3). The complete alignment, characterized by greater overall branch support, was used for ML-based phylogenetic reconstructions.

For both the UCE and *CYTB* datasets, we reconstructed phylogenetic relationships using maximum likelihood (ML) and Bayesian inference (BI) methods. These methods accommodate evolutionary complexity in different ways based on the input sequence alignments, thus ensuring robust comparative reconstructions of sequence relationships.

We conducted the ML analysis in IQ-TREE 3 (Minh et al., 2020; Wong et al., 2025), using the ModelFinder Plus function with the data partitioned by codon position. This approach instructs IQ-TREE to first select the optimal substitution model for each partition via ModelFinder and then proceed with the tree search, automatically merging similar models across partitions. Branch support was assessed using 5000 ultrafast bootstrap replicates with NNI optimization (Hoang et al., 2017) and SH-like approximate likelihood-ratio test replicates (Guindon et al., 2010). For clade support, we interpreted ultrafast bootstrap values as measures of resampling consistency under the model rather than as direct probabilities of clade correctness, and we primarily consider nodes that show concordant high ultrafast bootstrap and SH-aLRT support and are consistent with the corresponding Bayesian posterior probabilities.

The BI analysis was performed in BEAST v2.7.7 (Bouckaert et al., 2019) applying substitution models selected in ModelFinder (Kalyaanamoorthy et al., 2017), using BIC as a criterion (Schwarz, 1978). We implemented two independent runs, each constituting 200 million Markov chain Monte Carlo (MCMC) chains sampled at 20,000 intervals, with relaxed log-normal clock priors, which allow for rate variation among branches, and the Yule process speciation model, which assumes a constant speciation rate over time. The BI results were visualized in Tracer to assess sampling adequacy; effective sample sizes of 200 or higher were considered adequate. The log and tree files from the two runs were combined in LogCombiner (Rambaut and Drummond, 2024a) and TreeAnnotator 2.7.7 (Rambaut and Drummond, 2024b) to summarize trees and obtain a maximum clade credibility tree, with a 10% burn-in applied. The output trees were viewed and drawn either in FigTree (Rambaut, 2012) or tvBOT (Xie et al., 2023).

2.5 Genetic diversity, and species delimitation

We performed species delimitation using the *CYTB* alignment and the resulting ML and BI trees using multiple tree- and distance-based methods: the branch-cutting method (Mikula, 2018), the multirate Poisson tree processes algorithm (Kapli et al., 2017), the single-threshold general mixed Yule coalescent model (Fujisawa and Barraclough, 2013), and the assembly of species by automatic partitioning methods (Puillandre et al., 2021). The delimitations were implemented with the default settings for better output comparability. We resolved the differing number of species units suggested by different methods into a consensus clustering scheme of molecular operational taxonomic units (MOTUs) using LIMES (Ducasse et al., 2020).

We estimated evolutionary divergence within and among the delimited MOTUs using MEGA12 (Kumar et al., 2024), and various DNA polymorphism and divergence indices in DnaSP v6.12.03 (Rozas et al., 2017). We also examined genealogical relationships and geographical structuring of MOTUs using haplotype networks in PopART v1.7 (Bandelt et al., 1999; Leigh et al., 2015).

2.6 Morphological diversity

For a subset of samples, we integrated standardized external and craniodental morphometrics with the genetic MOTUs to generate species-identification hypotheses (Dayrat, 2005; Padial et al., 2010). Candidate species were based on latest checklist descriptions and original descriptions (Wilson et al., 2016; Mammal Diversity Database, 2025). External variables recorded in the field or transcribed from specimen tags comprised body mass, head–body length, tail length, ear length, and hindfoot length (Torre et al., 2004). We obtained cranial and dental measurements from cleaned skulls with digital calipers (0.01 mm resolution) using standard small-mammal craniometric landmarks (Carleton and Van der Straeten, 1997). The complete set of variables extracted included: greatest length of skull (GLS), condylo-incisive length (CI), zygomatic breadth (Zyg), braincase breadth (Brain), rostral/cranial height (Height), upper toothrow length (UTRL), lower toothrow length (LTR), lower molar row length (m1–m3), upper molar arcade breadth at M3 (m3–m3), and mandible length (Mand)—Table S1. Prior to comparisons and analyses, we screened for outliers in each MOTU using Tukey’s $1.5 \times \text{IQR}$ to handle potential errors arising from ontological and sex dimorphism. We used PCA to visualize whether MOTUs occupied distinct trait spaces and estimate assignment to candidate species (Jolliffe, 2011). Identifications were considered supported when key univariate measurements were concordant with documented species ranges and qualitative diagnoses in the checklist references (Dodge, 2008).

2.7 UCE species tree estimation

We reconstructed a species tree from the unfiltered UCE loci using the multi-species coalescent program ASTRAL v5.7.8 (Zhang et al., 2018). Maximum-likelihood gene trees were first inferred for every locus using IQ-TREE, employing ModelFinder for best-fit substitution models, and node support was assessed with 1,000 ultrafast bootstrap pseudoreplicates. The resulting tree files were then concatenated and supplied to ASTRAL, which optimizes the proportion of quartet relationships in the gene trees that agree with the species tree.

2.8 Divergence dating

To allow comparisons of topology and node age, we dated the divergences separately for the UCE and *CYTB* datasets. For the input datasets, we used the redundancy-filtered *CYTB* alignment. Lineage divergences were reconstructed under the Bayesian multispecies coalescent with StarBEAST3 in BEAST 2 (Douglas et al., 2022), a framework that samples gene trees and a species tree in a single Markov chain. We defined taxon sets to correspond to the consensus MOTUs to avoid arbitrary lumping of genetically cohesive lineages (Ducasse et al., 2020; Puillandre et al., 2024). To balance computational tractability and phylogenetic signal, we analyzed 372 loci that were selected from the top decile of the 90% taxon-occupancy UCE set based on parsimony-informative sites (Fig. S3). The *CYTB* alignment was analyzed as a single partition since uneven and extensive missing data would otherwise overparameterize a codon-position scheme and inflate variance (Kainer and Lanfear, 2015).

The divergence time estimates were constrained at two secondary calibration nodes (the family Gliridae and the African subfamily Graphiurinae) that synthesize recent molecular and fossil evidence (Montgelard et al., 2003; Nunome et al., 2007; Freudenthal and Martínez-Suárez, 2013); Petrova et al. (2024). For the family Gliridae node, we used the 95% highest-posterior-density (HPD) interval of 23.2–36.2 Ma reported for the family crown age in Álvarez-Carretero et al. (2022). This divergence was set in BEAST as a log-normal prior with offset = 0, $\mu = 3.367$, and $\sigma = 0.113$, which anchored the lower quantile (2.5%) at 23.2 Ma and upper quantile (97.5%) at 36.2 Ma, with a mean of 29.2 Ma. On the other hand, for the subfamily Graphiurinae node, we used the mitogenomic estimate of 16.64 Ma (HPD = 11.43–21.97 Ma) from Petrova et al. (2024), whose taxon sampling included the enigmatic *Selevinia* and refined Graphiurinae ages. We therefore used a log-normal prior with offset = 0, $\mu = 2.812$, and $\sigma = 0.146$, which yielded a median of 16.8 Ma, a lower (2.5%) quantile at 12.5 Ma, and an upper (97.5%) quantile at 22.2 Ma. These priors are broad because dormouse age at the family level remains contentious; they encompass earlier estimates that span late-Eocene to early-Miocene ages, such as the 50 Ma estimated by Montgelard et al. (2003), 55 Ma by Nunome et al. (2007), and 34–41 Ma by Bover et al. (2020), while accommodating paleontological proposals that placed a mid-Miocene ancestor at ~16 Ma (Freudenthal and Martínez-Suárez, 2013), thereby providing a coherent temporal framework.

We used an independent HKY + Γ (four Γ -distributed rate categories) substitution model and an uncorrelated log-normal relaxed clock, with topologies linked across loci. In addition, in the *CYTB* analysis, we specified an informative mean clock rate prior to 0.02 substitutions per site per million years, reflecting the $\approx 2\%$ Myr⁻¹ mitochondrial substitution rate typical for mammals (Bininda-Emonds, 2007). We ran two independent MCMC chains, each with 100 million generations, sampling every 10,000 generations. The log and tree output files were combined in LogCombiner after discarding the first 10% as burn-in. We inspected convergence in Tracer v1.7.2 (Rambaut et al., 2018) to confirm that all the parameters had effective sample sizes well above 200 and summarized the posterior log and tree distribution in TreeAnnotator.

2.9 Historical biogeography analysis

We inferred the biogeographic history of *Graphiurus* with BioGeoBEARS (Matzke, 2013, 2014) using the time-calibrated BEAST output, applying all six biogeographic models: DEC (dispersal-extinction-cladogenesis), DIVALIKE (dispersal-vicariance-like), and BAYAREALIKE (Bayesian-Like), plus their jump "+J" option to account for founder-effect speciation events (jump dispersal), which can improve model fit (Clark et al., 2008). Model fit was evaluated with AIC. To minimize geographic overgeneralization, we complemented the standard continent-wide partitions, sensu Linder et al. (2012), with seven paleoenvironmental macroregions derived from late Miocene to Holocene conditions' reconstructions. These regions were drawn to follow the main forest blocks, montane refugia and woodland belts that shifted with East African rifting, forest–savanna oscillations, and Holocene land-use changes (Voelker and Light, 2011; Cohen et al., 2022; Dommain et al., 2022; Dagallier et al., 2024), and included: (A) Albertine Rift complex: montane spine of the western Rift plus adjacent Congolian lowlands; (B) Angolan Highlands: discrete Afromontane archipelago on the Angolan escarpment; (C) Eastern Africa Highlands: the sky-island chain from the Kenyan and Tanzanian highlands through the Eastern Arc to northern

Malawi; (D) Ethiopian Highlands: montane forests, grasslands and moorlands above ~1,800 m on the Horn-of-Africa plateau; (E) South Africa (Southern Escarpment: the Great Escarpment of Zimbabwe–South Africa, including the Highveld outliers; (F) West Africa (Upper–Lower Guinea forests: Upper Guinea rainforest block across the Dahomey Gap to the Lower Guinea coastal forests; and (G) Zambezan Miombo belt: the Miombo woodland of the central–southern Zambezan region. These ‘regional’ biogeographic units capture refugial areas linked to long-term climatic stability (Couvreur et al., 2021; Huntley, 2023) and diversification hotspots such as the Ethiopian massif and southern escarpment (Frost, 1996; Pennington et al., 2004; Clark et al., 2011; Cuypers et al., 2022; Behrends et al., 2024; Taylor et al., 2024). Incorporating these boundaries therefore maximizes the likelihood that inferred ancestral ranges track the actual corridors and barriers that shaped *Graphiurus* diversification, rather than historical coarse biogeographic boundaries.

3 Results

3.1 *CYTB* phylogeny, diversity, and species limits

In the Gliridae family *CYTB* dataset, *Graphiurus* was strongly supported as a monophyletic early-branching lineage (Fig. S4). The divergence between *Graphiurus* and other genera based on Kimura two-parameter (K2P) distances ranged between 25.4% (*Glis*) and 30.8% (*Eliomys*), while within the family ranged between 23.1% (*Glis* vs. *Muscardinus*) and 32.3% (*Muscardinus* 370 vs. *Selevinia*), with *Graphiurus* (10.2%) and *Dryomys* (14.9%) having the highest internal diversity (Table S4).

Within *Graphiurus*, species delimitation using the *CYTB* dataset identified over 35 taxonomic units with different splitting topologies between methods (Fig. 3). Based on mammalian classification frameworks which generally consider *CYTB* divergences of <2% as indicative of intraspecific variation and divergences >5% as suggestive of lineages approaching or at species boundaries, with sister species typically exhibiting divergences around 10–13% (Bradley and Baker, 2001; Baker and Bradley, 2006), genetic distances within and between units, along with geographic consistency, we identified 24 strongly monophyletic units, which we classified as species-level MOTUs for subsequent analyses. These MOTUs exhibited pairwise K2P distances between 3.2% and 32.4%, with overall higher between-MOTU distances than within-MOTU distances (Table 1). They also exhibited distinct geographical structuring (Fig. 2B; Fig. S5) and were strongly supported, on average, in both the maximum likelihood (ML) and Bayesian inference (BI) analyses (Fig. 3, Fig. S6 & S7). Given the widespread use and robustness of ML methods, we use the ML tree as the primary reference for phylogenetic results.

Two major groups were recovered in the *Graphiurus* genus. One group is comprised of three early diverging MOTUs (MOTU1[*nagtglassii*], MOTU2, and MOTU3), which are highly differentiated from the rest of the tree and from each other, forming a monophyletic West African group (BS=99%; PP=100%) restricted to the Upper Guinea forests (Fig. 2B; Fig. S5). The genetic divergence of these three MOTUs from all others and from each other is notably large—K2P

distance: 22% to 32%—comparable to the divergences observed among genera within Gliridae (Fig. S4 and Table S4). External body, craniodental measurements (Table S1; Fig. S8), and sampling overlap to type locality (Fig. 2A) link MOTU1[*nagtglasii*] to Nagtglas's African Dormouse (*Graphiurus nagtglasii* Jentink, 1888), while we could not link the remaining two MOTUs to any available names.

The second major group is comprised of all remaining samples in a highly divergent, monophyletic clade (BS=100%; PP=100%), widely distributed throughout sub-Saharan Africa (Fig. 3; Fig. 2B–D). The earliest-diverging MOTU within this widely distributed group, MOTU4, is represented by a single sample from northeastern South Africa at Lajuma, followed by a group (BS=66%; PP=100%) of two sister MOTUs: MOTU5 from the lower Guinea represented by samples from the Central African Republic (Dzanga-Sangha) and Gabon (Moukalaba-Doudau) and MOTU6 from the Eastern Arc Mountains (East Usambara, northeastern Tanzania) and northern Malawi's Misuku Hills (Fig. 2; Fig. S5). The subsequent divergence, MOTU8, is an East African clade in the arid Somali–Massai savannas in northern Tanzania, Kenya, and the Afar triangle in Ethiopia, with Ethiopian samples separated from the Kenya and Tanzania clusters (Fig. 3, Fig. S5). This is followed by a monophyletic group (BS=77%; PP=90%) of two very geographically distinct populations: MOTU7 (Angolan escarpment (Tundavala)) and its sister, MOTU9, an Upper Guinea clade syntopic with *nagtglasii* in the Wologizi-Wonegizi-Ziama forest landscape.

The rest of the *Graphiurus* tree is a geographically wide-ranging group of 15 MOTUs across the Afrotropical realm, bifurcating into two main subgroups: (i) a dense Eastern Afromontane group of 13 MOTUs (BS=51%; PP=100%) and (ii) a two-MOTU group (BS=100%; PP=100%) abutting the Sudanian Savanna from the Ethiopian Highlands to the Upper Guinean forests. The latter further splits into two strongly supported, reciprocally monophyletic MOTUs: MOTU11[cf. *lorraineus*] containing Benin, Guinea, and Côte d'Ivoire samples, and MOTU12 containing Ethiopia, the Democratic Republic of the Congo, Cameroon, Equatorial Guinea, and Nigeria samples. External body and craniodental measurements (Table S1), and sampling overlap to type locality, link MOTU11[cf. *lorraineus*] to Lorraine's African Dormouse *Graphiurus lorraineus* Dollman, 1910. We therefore call it *G. cf. lorraineus* awaiting formal taxonomic evaluation. The southwestern Angola-described *G. kelleni* (Reuvsen, 1890) is one of the most likely names for MOTU12. However, the notable sampling gap between our samples and this type locality, and the emerging strong clade-to-geography relationships make it difficult to confidently associate it without morphological backup. All delimitation methods split MOTU11[cf. *lorraineus*] and MOTU12 into several units (Fig. 3), with haplotype networks showing strong internal geographical structuring (Fig. S5).

Among the 15 MOTUs, the unit we designate as MOTU24[cf. *microtis*] is recovered as the most internally nested, strongly supported monophyletic clade (BS=100%; PP=100%) (Fig. 1; Fig. 2). Despite spanning a broad region across East Africa, this MOTU shows little internal phylogeographic structuring (Fig. 2; Fig. S5). We tentatively interpret this MOTU as a *species indeterminabilis* of Noack's African dormouse, *Graphiurus microtis* (Noack, 1887), hereafter MOTU24[cf. *microtis*], based on its wide distribution, which overlaps the type locality of *G.*

microtis in southwestern Democratic Republic of the Congo, and its generally broad morphotypes (Fig. S8).

The remaining 12 Eastern Afromontane MOTUs form a grade between MOTU24[cf. *microtis*] and the MOTU11[cf. *lorraineus*] + MOTU12 clade, arranged as a stepwise series of weakly supported nested branches (Fig. 1; Fig. 2). Each MOTU is itself recovered as a well-supported monophyletic group, but relationships among them are mostly unresolved (BS=44%–66 %), except for the crown-ward sequence—(((MOTU24[cf. *microtis*], MOTU23), MOTU22), MOTU21). MOTUs are also broadly congruent in their geographic distributions (Fig. 1; Fig. 2): MOTU22 spans northern Kenya and extends into western and central Ethiopia; MOTU21 tracks the Eastern Arc from the Shimba Hills (southern Kenya) south to the Rungwe–Misuku highlands (southern Tanzania/northern Malawi); MOTU19 and MOTU20 are restricted to the Albertine Rift. MOTU17 is represented by samples from Hombe in southwestern Angola; MOTU16 occurs in the central Kenyan highlands (Mount Kenya and the Aberdare Range); MOTU10 appears syntopic with MOTU21 in the southern Eastern Arc, specifically the Rungwe Highlands (southwestern Tanzania) and Misuku Hills (northern Malawi) (Fig. 2; Fig. S5); and MOTU15 and MOTU18 are represented in our southernmost samples, with the former represented by material from the Molopo region of northern South Africa, and the later from Bulembu in northwestern Eswatini.

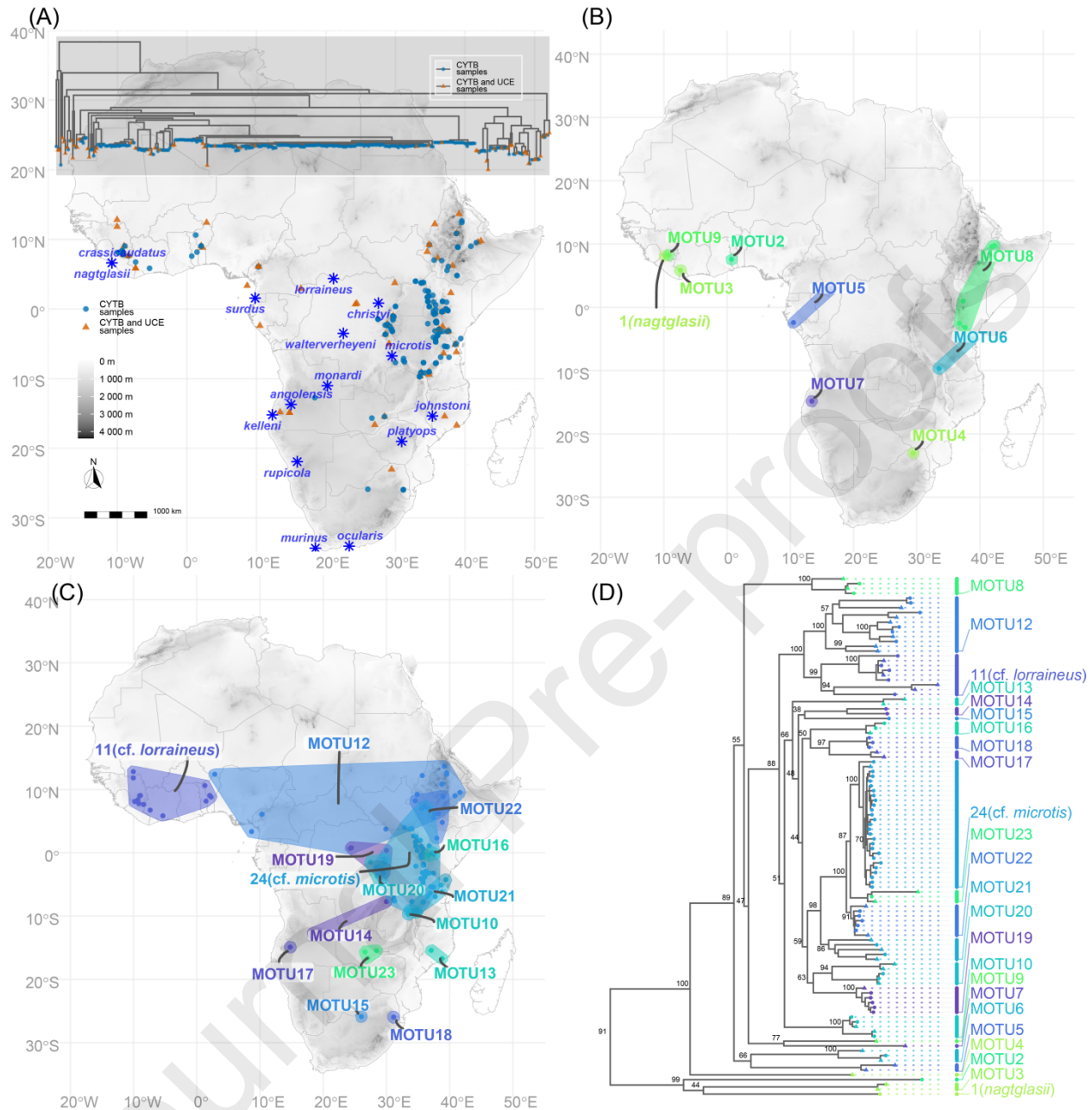


Fig. 2. Sampling and cytochrome b gene (*CYTB*) phylogeny of samples used in the study. (A) Geographical distribution of the *Graphiurus* samples analysed, all of which were genotyped for *CYTB*; symbol colour and shape indicate whether samples were sequenced at *CYTB* only or at both *CYTB* and ultraconserved-element (UCE) for each locality and correspond to the colour scheme in the simplified *CYTB* phylogeny shown inset above the map. Type localities of currently accepted *Graphiurus* species (Wilson et al., 2017) are marked for reference (blue star shapes). (B) Distribution and genetic assignment of samples assigned to nine early-diverging MOTUs. (C) Distribution and genetic assignment of samples belonging to the 15 Plio-Pleistocene fast-radiating MOTUs. (D) The *CYTB*-reconstructed phylogenetic relationships within *Graphiurus* with clade labels and tip shape colors matching those in B and C.

Table 1 Genetic distances (Kimura 2-Parameter (K2P) distances) between the recovered molecular operational taxonomic units (MOTUs) inferred from estimates of average evolutionary divergence over sequence pairs. The number of base differences per site from averaging over sequence pairs between MOTUs are shown as proportions in the lower diagonal and the corresponding standard error estimates in the upper diagonal. The diagonal reports the corresponding within-MOTU distances. The analysis included 331 mitochondrial cytochrome b gene (*CYTB*) sequences with the pairwise deletion option applied to ambiguous positions. See Table S5 for the corresponding DnaSP-derived genetic diversity indices. n/c = Not computed; for MOTUs that had <2 samples.

MOTU	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1[<i>nagtglasii</i>]	n/c	0.021	0.021	0.025	0.029	0.022	0.024	0.025	0.041	0.026	0.022	0.020	0.023	0.026	0.030	0.022	0.026	0.023	0.024	0.023	0.021	0.022	0.026	0.021
MOTU2	0.248	n/c	0.022	0.023	0.024	0.022	0.026	0.025	0.043	0.025	0.024	0.022	0.024	0.027	0.031	0.027	0.033	0.029	0.024	0.024	0.023	0.025	0.027	0.024
MOTU3	0.204	0.237	0.01	0.023	0.033	0.022	0.033	0.035	0.046	0.030	0.028	0.024	0.031	0.029	0.026	0.030	0.034	0.027	0.028	0.027	0.026	0.029	0.030	0.026
MOTU4	0.279	0.275	0.251	n/c	0.019	0.018	0.023	0.023	0.027	0.019	0.019	0.016	0.024	0.021	0.022	0.021	0.022	0.019	0.019	0.017	0.018	0.019	0.023	0.018
MOTU5	0.292	0.244	0.269	0.191	0.03	0.019	0.021	0.020	0.030	0.020	0.019	0.018	0.023	0.021	0.025	0.017	0.023	0.021	0.019	0.019	0.020	0.022	0.023	0.023
MOTU6	0.269	0.277	0.237	0.215	0.184	0.05	0.021	0.018	0.028	0.019	0.016	0.014	0.019	0.017	0.019	0.016	0.019	0.016	0.016	0.017	0.015	0.014	0.016	0.015
MOTU7	0.270	0.263	0.297	0.225	0.184	0.215	n/c	0.017	0.024	0.020	0.021	0.019	0.019	0.021	0.026	0.020	0.021	0.024	0.023	0.023	0.023	0.022	0.022	0.021
MOTU8	0.248	0.273	0.285	0.216	0.184	0.178	0.183	0.03	0.021	0.015	0.015	0.017	0.018	0.014	0.020	0.014	0.017	0.017	0.016	0.015	0.014	0.015	0.020	0.015
MOTU9	0.259	0.290	0.257	0.163	0.203	0.175	0.135	0.123	n/c	0.026	0.024	0.024	0.025	0.025	0.023	0.026	0.027	0.028	0.023	0.023	0.023	0.023	0.022	0.023
MOTU10	0.292	0.303	0.282	0.210	0.196	0.201	0.194	0.138	0.140	0.02	0.014	0.014	0.016	0.014	0.016	0.013	0.017	0.015	0.014	0.012	0.015	0.014	0.018	0.014

11[cf. <i>lorraineus</i>]	0.276	0.290	0.278	0.211	0.178	0.186	0.206	0.154	0.158	0.135	0.07	0.010	0.015	0.015	0.015	0.012	0.015	0.014	0.012	0.013	0.013	0.013	0.016	0.014
MOTU12	0.276	0.282	0.273	0.217	0.189	0.197	0.204	0.165	0.162	0.147	0.127	0.08	0.013	0.015	0.016	0.012	0.016	0.011	0.012	0.013	0.013	0.013	0.015	0.013
MOTU13	0.251	0.258	0.267	0.229	0.195	0.215	0.180	0.174	0.167	0.139	0.140	0.140	0.02	0.017	0.018	0.017	0.016	0.014	0.014	0.015	0.015	0.014	0.017	0.014
MOTU14	0.301	0.298	0.263	0.232	0.181	0.180	0.210	0.152	0.140	0.136	0.138	0.162	0.158	0.07	0.015	0.013	0.012	0.013	0.015	0.013	0.013	0.013	0.016	0.013
MOTU15	0.295	0.322	0.239	0.192	0.192	0.177	0.211	0.160	0.125	0.119	0.136	0.152	0.141	0.130	n/c	0.016	0.015	0.014	0.013	0.015	0.014	0.014	0.015	0.014
MOTU16	0.269	0.289	0.281	0.230	0.160	0.188	0.202	0.157	0.148	0.134	0.133	0.140	0.142	0.132	0.156	0.01	0.013	0.011	0.011	0.011	0.011	0.011	0.014	0.012
MOTU17	0.252	0.300	0.277	0.213	0.202	0.179	0.213	0.166	0.144	0.142	0.146	0.157	0.145	0.120	0.131	0.117	0.02	0.010	0.013	0.015	0.012	0.014	0.015	0.012
MOTU18	0.266	0.324	0.248	0.218	0.177	0.179	0.210	0.162	0.146	0.140	0.129	0.146	0.135	0.125	0.119	0.112	0.083	0.01	0.011	0.013	0.011	0.010	0.013	0.010
MOTU19	0.292	0.304	0.275	0.205	0.186	0.192	0.212	0.159	0.116	0.129	0.121	0.151	0.119	0.127	0.107	0.119	0.106	0.104	0.01	0.011	0.011	0.011	0.014	0.012
MOTU20	0.295	0.291	0.262	0.210	0.198	0.188	0.214	0.162	0.148	0.133	0.141	0.162	0.143	0.138	0.141	0.129	0.140	0.134	0.110	0.05	0.010	0.010	0.012	0.010
MOTU21	0.271	0.290	0.257	0.208	0.195	0.181	0.195	0.158	0.138	0.136	0.138	0.155	0.137	0.137	0.122	0.125	0.121	0.112	0.111	0.118	0.06	0.008	0.010	0.008
MOTU22	0.250	0.295	0.263	0.207	0.192	0.160	0.194	0.149	0.122	0.116	0.119	0.142	0.133	0.127	0.129	0.107	0.111	0.096	0.100	0.105	0.085	0.01	0.007	0.005
MOTU23	0.265	0.301	0.253	0.231	0.200	0.174	0.196	0.192	0.116	0.135	0.134	0.158	0.137	0.146	0.137	0.130	0.124	0.111	0.118	0.124	0.099	0.048	0.03	0.009

24[cf. *microtis*] 0.254 0.296 0.259 0.218 0.207 0.165 0.193 0.162 0.127 0.119 0.125 0.145 0.130 0.128 0.127 0.114 0.110 0.104 0.107 0.108 0.089 0.032 0.054 0.01

3.2 UCE phylogeny and its comparison with the *CYTB* tree

We were unable to obtain UCE representative samples for five of the *CYTB* MOTUs (MOTU1[*nagtglasii*], MOTU2, MOTU15, MOTU18, and MOTU14); for these, we rely on the *CYTB*-only units for all inferences.

The UCE phylogeny (Fig. 3B) largely corroborates the *CYTB* tree (Fig. 3B, Fig. S9), while refining MOTU boundaries and arrangement among major lineages. Several relationships that were only moderately supported or ambiguously placed in the *CYTB* tree were fully resolved in the UCE tree, with consistently high node support (ML BS $\geq 96\%$; BI PP=1.0) across trees (Fig. 2; Fig. 3).

The three non-graphiurine species form a clade, with *Glis glis* sister to (*Dryomys nitedula* + *Muscardinus avellanarius*). Within the genus *Graphiurus*, the same set of primary MOTUs is recovered as in the *CYTB* phylogeny (Fig. 3), with the positions of the early-diverging lineages being comparable and the two nested, broad geographic groupings evident in each (Fig. 2; Fig. 3; Fig. S6).

The deep West African lineage MOTUs and the earliest-diverging *Graphiurus* MOTUs retained the same backbone relationships in the UCE phylogeny, with differences only in split times; in order of successive splits: MOTU3 $\rightarrow$ MOTU4 $\rightarrow$ MOTU6 $\rightarrow$ MOTU5 $\rightarrow$ MOTU7 $\rightarrow$ MOTU8 $\rightarrow$ MOTU9.

Beyond the backbone, the remaining radiation resolved into two main groups: a Zambezian–Eastern Afromontane assemblage spanning the Angolan highlands to the Ethiopian highlands (MOTU17, (MOTU10, MOTU13), (MOTU21, (MOTU22, (MOTU23, MOTU24[cf. *microtis*])))) and an Albertine–Kenya–Ethiopia-to-Upper Guinea assemblage wherein the Albertine Rift lineages branch basally to a group consisting of Kenya + Ethiopia-to-Upper Guinea MOTUs ((MOTU20 + MOTU19), (MOTU16, (MOTU11[cf. *lorraineus*], MOTU12))) (Fig. 2; Fig. 3).

The weakly supported *CYTB* grade between MOTU24[cf. *microtis*] and the Ethiopia–Congo–Guinea lineage collapses to different, more strongly supported topologies in the UCE tree. Conversely, some *CYTB*-defined relationships are not recovered in UCE (e.g., the *CYTB* grouping that links MOTU7 with MOTU9 and MOTU5 with MOTU6). In contrast, the two Albertine Rift lineages distinct in the *CYTB* tree (MOTU19 and MOTU20) form a single MOTU in the UCE tree. Within the Zambezian–Eastern Afromontane assemblage, MOTU10 is a sister to MOTU13; together, they form a clade with MOTU17, unlike in the *CYTB* tree, where this relationship is not recovered.

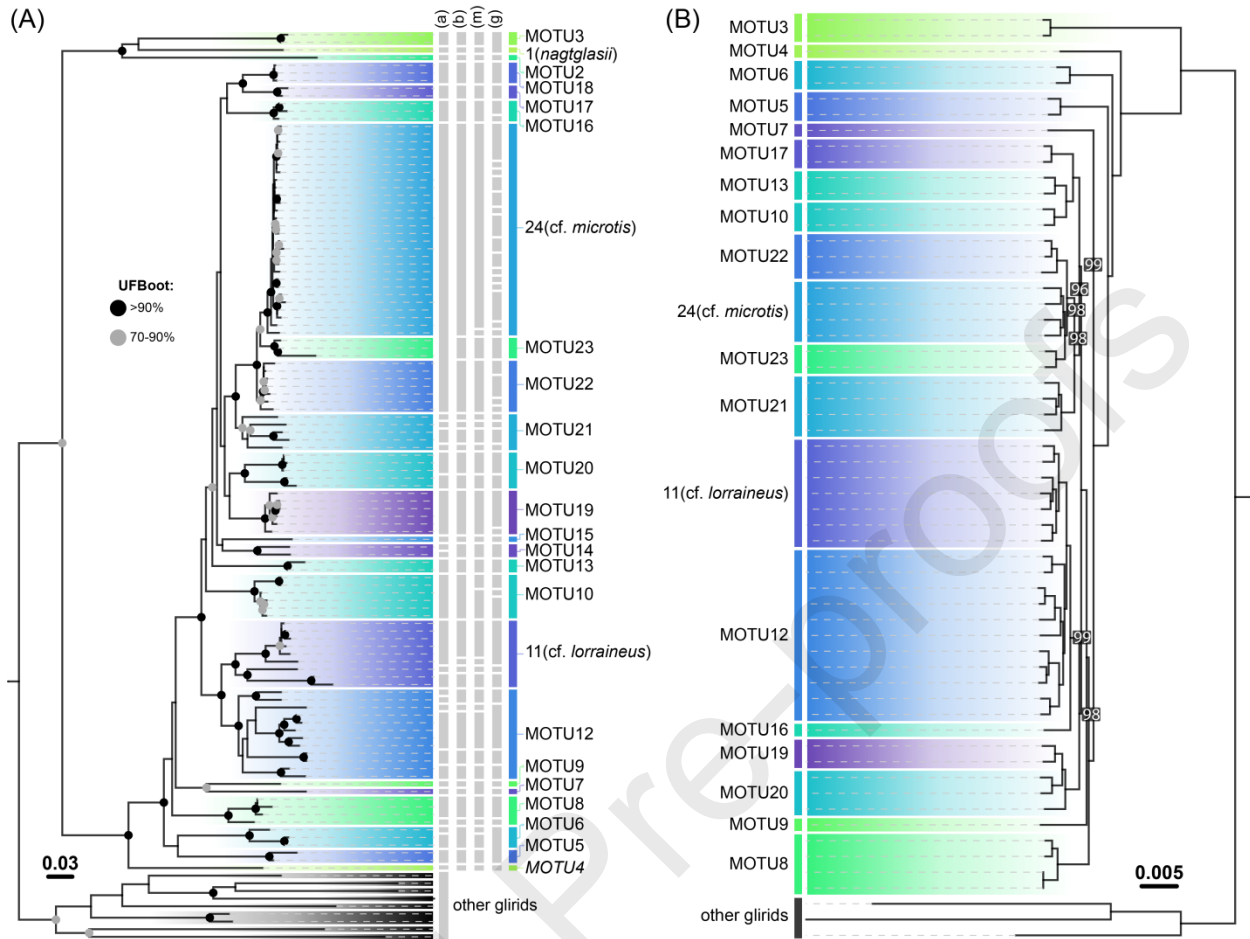


Fig. 3. Phylogenetic relationships of molecular operational units (MOTUs) within the African dormouse genus *Graphiurus*, inferred from mitochondrial cytochrome b (*CYTB*) sequences and genome-wide ultraconserved elements (UCEs). Panel A shows the maximum-likelihood (ML) tree reconstructed from *CYTB* sequences. Greyscale blocks indicate partitions proposed by individual species-delimitation methods [(a) the assembly of species by automatic partitioning methods (Puillandre et al., 2021), (b) the branch-cutting method (Mikula, 2018), (g) the single-threshold general mixed Yule coalescent model (Fujisawa and Barraclough, 2013), and (m) the multirate Poisson tree processes algorithm (Kapli et al., 2017)]; the final bars highlights the consensus 24-MOTU partition adopted as putative species units, which correspond to node blocks shaded in matching colours. UFBboot = bootstrap support. See Fig. S7 for the full *CYTB* tree. Panel B shows the ML tree from 4,452 genome-wide ultraconserved elements (UCEs) loci annotated with the same *CYTB*-derived MOTU identifications for comparison; branch labels show bootstrap support values that were <100%. The UCE tree lacks five MOTUs from the *CYTB* tree, for which we were unable to obtain samples.

3.3 Divergence dating estimates

The UCE dataset indicated that diversification within Gliridae began at 30 Ma (HPD 35.3–24.9 Ma), marked by the split between the ancestor of *Graphiurus* and the ancestor of the remaining non-graphiurine genera (Fig. 4). Near-contemporaneously, the Glirinae representative *Glis glis* diverged from the Leithiinae lineage represented here by (*Muscardinus avellanarius* + *Dryomys nitedula*), and these two leithiines subsequently split in the mid-Oligocene at 26.3 Ma (HPD 31.1–21.9 Ma). In contrast, the *CYTB* dataset suggested a younger onset of divergence within Gliridae,

at 25.4 Ma (HPD 30.3–20.6 Ma), again corresponding to the split between the *Graphiurus* ancestor and the ancestor of all other genera (Fig. S4). Diversification within these non-graphiurine lineages commenced at 20.8 Ma (HPD 25.95–16.0 Ma) with the split between Leithiinae and Glirinae.

Diversification within Graphiurinae was dated to 19.3 Ma (HPD 23.2–15.8 Ma) under *CYTB* (Fig. 4B); however, in the UCE dataset it was dated to the middle Miocene at 14.7 Ma (HPD 17.4–12.2 Ma), corresponding to the split between the representative MOTU from the West African deep lineage (MOTU3) and a lineage comprising all other MOTUs (the widespread sub-Saharan African MOTUs) (Fig. 4A). Following this MOTU3 split, no additional splits are inferred within the widespread sub-Saharan African lineage until the late Miocene, when early-diverging MOTUs arose sequentially: MOTU4 at 7.9 Ma (HPD 9.4–6.6 Ma), MOTU6 at 7.2 Ma (HPD 8.6–6.0 Ma), MOTU5 at 6.8 Ma (HPD 8.1–5.7 Ma), MOTU7 at 5.6 Ma (HPD 6.6–4.6 Ma), MOTU8 at 5.1 Ma (HPD 6.0–4.2 Ma), and MOTU9 at 4.9 Ma (HPD 5.7–4.0 Ma) (Fig. 4, Table S6). The next split at 4.5 Ma (HPD 5.2–3.7 Ma) divided the sister clade to MOTU9—containing the remainder of Graphiurinae—into two sister lineages (Fig. 4, Table S6).

In the first of these lineages, the Albertine Rift clade (MOTU19 + MOTU20) branched off at 4.2 Ma (HPD 5.0–3.5 Ma), followed by the central-Kenyan MOTU16 at 3.3 Ma (HPD 3.9–2.7 Ma), and subsequently the Ethiopia–Upper Guinean pair (MOTU11[cf. *lorraineus*] + MOTU12), which split at 2.9 Ma (HPD 3.4–2.4 Ma) (Fig. 4). In the second lineage, two primary groups—(MOTU10 + MOTU13 + MOTU17) versus (MOTU21 + MOTU22 + MOTU23 + MOTU24[cf. *microtis*])—diverged at 4.2 Ma (HPD 4.9–3.4 Ma), followed by near-synchronous internal diversification onset at the late early Pliocene/early mid-Pliocene transition: 3.5 Ma (HPD 4.1–2.9 Ma) and 3.7 Ma (HPD 4.3–3.0 Ma), respectively (Fig. 4).

All corresponding divergences were recovered as earlier in time under *CYTB*. Among the central to southern Africa MOTUs for which we did not have UCE data, MOTU14 split from MOTU15 at 6.4 Ma (HPD 8.8–4.3 Ma), and MOTU18 split from MOTU17 at 3.7 Ma (HPD 5.3–2.4 Ma).

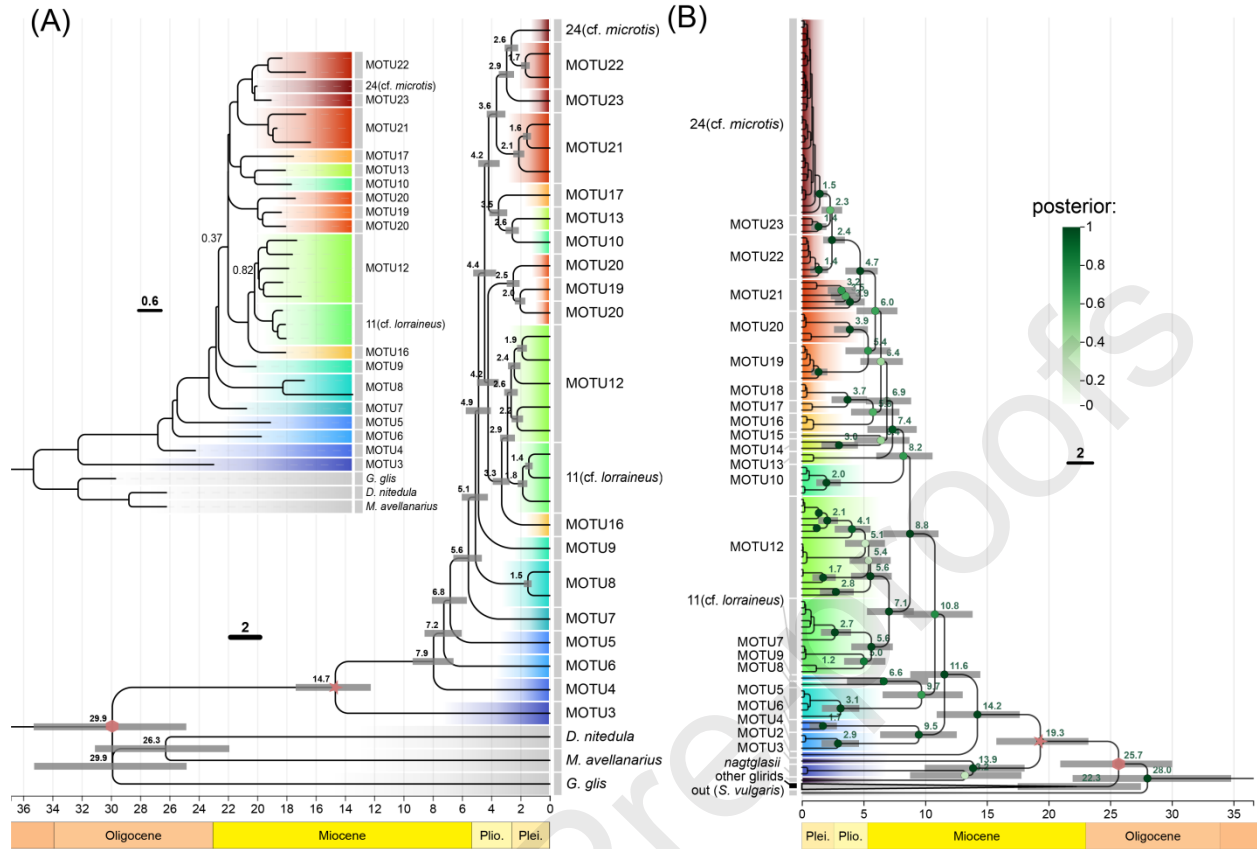


Fig. 4. Dating divergence of genus *Graphiurus*. (A) The tree recovered from ultraconserved elements, with an inset showing an ASTRAL species tree derived from the same dataset. The branch labels indicate instances where support values were less than 95%. (B) A comparative time-calibrated mitochondrial cytochrome b (*CYTB*) phylogeny, where the colors of node shapes represent posterior support values. Divergence times were based on secondary calibration points (light-red hexagon shape: the most recent common ancestor (MRCA) of the Gliridae family, and light-red star shape: the MRCA of the subfamily Graphiurinae). The x-axis represents time before present (in millions of years), while the grey node bars illustrate the 95% highest posterior density intervals surrounding the mean node ages, which are shown as node labels. Tip labels and greyscale blocks delineate the boundaries of the 24 molecular operational taxonomic units identified in this study. The UCE tree lacks five MOTUs from the *CYTB* tree, for which we were unable to obtain samples.

3.4 Ancestral area reconstruction

Comparisons between the continental biome states and the regional states showed that the latter yielded more biologically informative reconstructions of ancestral biogeography (Table S7). Consequently, we report ancestral area reconstructions based on these regionally defined states. Model selection favored parameterizations that included jump dispersal (+J), consistent with founder-event speciation (Table S7). The optimal UCE model, DIVALIKE+J, estimated near-zero traditional (anagenetic) dispersal and extinction rates but a nonzero jump parameter. This pattern was similar under the optimal *CYTB* model, DEC+J: parameter estimates converged on very low levels of anagenetic dispersal and effectively zero extinction while consistently maintaining nonzero founder-event estimates.

The ancestral state of Graphiurinae, and the inferred geographic context for the earliest within-continent history, were not resolved in the UCE tree (Fig. 5) but were resolved to a West African state in the *CYTB* tree (Fig. S10). The onset of divergence within the continent also remains ambiguous: a combined West Africa + East Africa range is most likely, with slightly less support for an entirely East Africa range. The first internal divergence within the genus *Graphiurus*, when the West African lineage (represented by MOTU3) split, was not resolved among the Eastern Afromontane, South Africa, and West Africa states (Fig. 5; Fig. S10). Even in the *CYTB* tree—where the highest-likelihood reconstruction is West Africa—Eastern Africa retained a similar likelihood (Fig. S10).

Across both datasets, the ancestral state along the backbone of early divergences within the pan-Afrotropical lineage (giving rise to the first four MOTUs) is consistently reconstructed as the Eastern African highlands. From this source area, the common ancestor of the Ethiopia-to-Upper Guinea group originated, including the western radiation, MOTU11[cf. *lorraineus*], and its sister (MOTU12). The crown of MOTU12 is unambiguously reconstructed as the Ethiopian highlands, whereas MOTU11[cf. *lorraineus*] is reconstructed as West Africa (Fig. 5; Fig. S10).

The eastern-to-southern/central radiations, including MOTU13, MOTU17, MOTU7, the Albertine Rift group (MOTU19 + MOTU20), and the remaining derived West Africa MOTUs (MOTU5 and MOTU9), also share ancestral states in the Eastern African highlands. Each regional MOTU then shows a single-area crown node matching its tip state, with divergences among tip MOTUs being predominantly single-area (Fig. 5; Fig. S10).

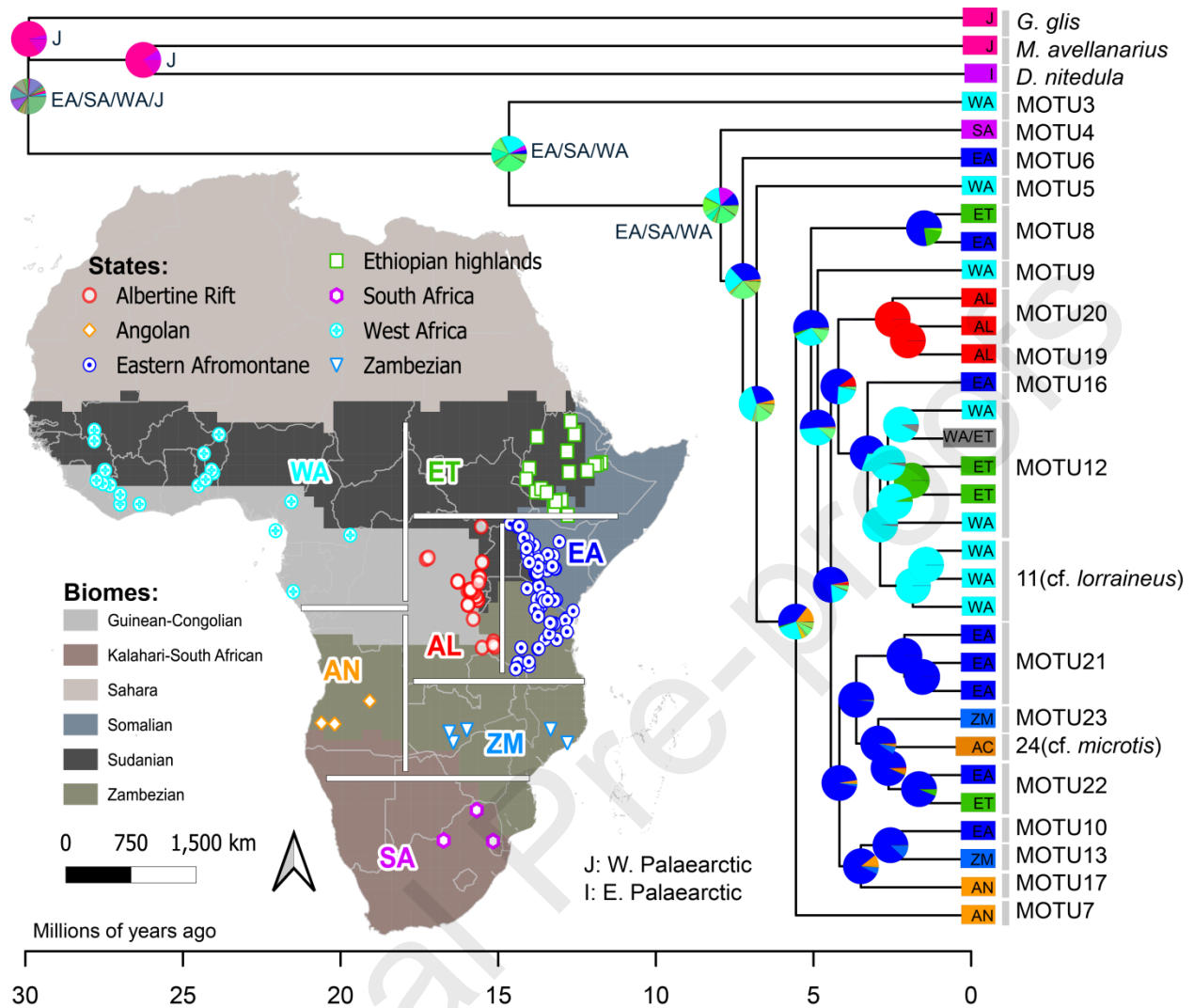


Fig. 5. Historical biogeographic reconstruction of genus *Graphiurus*. Ancestral ranges were inferred using the ultra-conserved elements' dataset and are shown as pie charts at major nodes on the time-calibrated tree. Pie-slice colors match the ancestral states on the map with proportions indicating their estimated probabilities. The map plots sampling localities grouped into the discrete clusters used as ancestral states, while the grey background indicates broader Afrotropical biomes. Tip labels mark the molecular operational taxonomic units recovered as species-level clades in the study. See the reconstruction based on the cytochrome b (*CYTb*) dataset in Fig. S10.

4 Discussion

The morphology-based taxonomy and distribution records of African dormice of the genus *Graphiurus* have long lacked a robust phylogenetic framework, obscuring the genus's evolutionary history and biogeography. Here we present the first comprehensive genetic dataset compiled for this group, which combines the mitochondrial cytochrome b gene and thousands of genome-wide ultraconserved elements. This integrative evidence enabled us to test various hypotheses regarding the origin and dispersal history of *Graphiurus*, a late Miocene origin, pulses

of diversification during the Plio–Pleistocene period, and geographically structured dispersal events. Summarized, our results show that an early-branching West African group forms a well-supported monophyletic lineage within genus *Graphiurus*, with genetic distances and range dynamics indicating that they might represent a distinct genus. Divergence-time estimates based on the UCE and *CYTB* datasets reveal that the ancestor of *Graphiurus* split from a Eurasian sister between the early Oligocene and late Oligocene, whereas most intra-African speciation occurred during the Plio–Pleistocene, coinciding with climatic oscillations that have been shown to have fragmented tropical forests and promoted allopatric speciation. These findings provide key new insight into a rarer, deeper Miocene split in an African rodent radiation, while echoing recent continent-wide syntheses documenting Plio–Pleistocene bursts of rodent diversification. We recover several deeply divergent species-level MOTUs whose phylogenetic positions do not straightforwardly link with available recognized names in this genus, highlighting the need for a thorough taxonomic revision of the genus to resolve morphological-range-genetic congruences and define more stable species boundaries.

4.1 Overview of *Graphiurus* phylogenetics

Our combined UCE plus *CYTB* phylogenies corroborate earlier cranial-morphological studies that placed the genus *Graphiurus* (subfamily Graphiurinae) outside Leithiinae + Glirinae radiation, suggesting an ancient split within dormice (Wahlert et al., 1993; Pickford and Morales, 1994; Bentz and Montgelard, 1999; Montgelard et al., 2003; Nunome et al., 2007). Limited-locus molecular sampling in early Gliridae work had hinted at this basal position but lacked genus-wide taxon coverage and comprehensive locus depth needed for decisive support (Bentz and Montgelard, 1999). By integrating thousands of UCE loci, we resolve the same topology with maximal support, mirroring the family-level trees produced from the same individuals and reinforcing a scenario in which Graphiurinae likely diverged in Eurasia before colonizing sub-Saharan Africa (Wahlert et al., 1993; Montgelard et al., 2003; Nunome et al., 2007). The delimited 24 MOTUs is a conservative partitioning scheme likely to change in the future in broader combinations with morphology and geography to define taxonomic boundaries between parapatric and sympatric populations. Still, when compared with the 15 currently formally recognized *Graphiurus* species (Wilson and Reeder, 2005; Happold, 2013; Monadjem et al., 2015; Wilson et al., 2017), there is an apparent shortfall in the recognized diversity of the genus.

The genetic distance between the two deep, reciprocally monophyletic lineages within *Graphiurus* that split in the middle Miocene (i) a West African lineage that lies outside all other *Graphiurus* and matches the historical subgenus *Aethoglis* (Pavlinov and Potapova, 2003), and (ii) a pan-African lineage containing the remaining MOTUs) equal or exceed those separating recognized small-mammal genera (Bradley and Baker, 2001; Baker and Bradley, 2006). The three MOTUs of the West African lineage (MOTU1[*nagtglasii*], MOTU2, and MOTU3) represent an evolutionarily cohesive West African radiation. Here, we could morphologically associate only one MOTU1[*nagtglasii*] with the subgenus *Aethoglis* type species *G. nagtglasii* (see measurements in Table S1). Historical accounts recognize *G. nagtglasii* as distinguished from subgenus *Claviglis* (e.g., *G. crassicaudatus*) by a larger body size, cranial proportions, and a distinctive hystricomorphous zygomasseteric (Holden, 1996; Pavlinov and Potapova, 2003;

Wilson and Reeder, 2005; Holden and Levine, 2009; Wilson et al., 2017). Although the recovered phylogenetic arrangement would justify elevating *Aethoglis* to full generic rank, to abide by the principle of typification, while avoiding taxonomic inflation based solely on genetic distances or geographic structure (Carstens et al., 2013; Braby et al., 2024), any formal nomenclatural changes are reserved for future studies that employ a coordinated morphological–genomic review of type material.

The earliest-diverging MOTUs in the pan-African lineage comprise several deeply isolated early-splitting MOTUs, suggesting a strong geographic imprint on lineage splitting. The South African branch (MOTU4) lies far outside the remaining 17 ingroup MOTUs, and the subsequent five early offshoots are each recovered as distinct, geographically restricted MOTUs. Its divergence from other MOTUs (Table 1) exceeds the generic thresholds proposed for rodents (~7–9% *CYTB*; Bradley and Baker (2001); Baker and Bradley (2006)), highlighting the need for targeted sampling in the Cape Fold Belt to test whether MOTU4 represents a single-species lineage (as in *G. platyops*, whose type locality is ~480 km from where the MOTU4 representative was obtained) or the stem of a broader southern radiation whose remaining lineages have yet to be sampled. The subsequent divergences following MOTU4—MOTU5, MOTU6, MOTU7, MOTU8, and MOTU9—are characteristic of species units that are evolutionarily well isolated and have relatively restricted ranges (Smyčka et al., 2023; Alzate et al., 2025). We expect these lineages to be readily diagnosable from later-emerging sympatric or syntopic species using morphological or molecular characters (Lukhtanov, 2019). Among these 17 remaining MOTUs, the lack of morphological diagnosis renders assigning them to available names impossible, reiterating the need for a complete genus taxonomic revision to link genetic MOTUs to morphological species. Such an effort would also help refine the higher-level taxonomy implicit in the deep splits between the earliest diverging MOTUs and the Plio-Pleistocene diversification burst.

Among the late-diverging MOTUs in the pan-African lineage, the first group to emerge comprises samples from the Albertine Rift (MOTU19 + MOTU20), Central Kenya (MOTU16), and Ethiopia-to-Upper Guinea (MOTU24[cf. *lorraineus*] + MOTU12). The Albertine Rift MOTUs were recovered as an early-splitting group, sister to the Central Kenya MOTU and the Ethiopia-to-Upper Guinea group. While they split into their respective monophyletic MOTUs at *CYTB*, they collapse into a single clade in the UCE tree. This discordance seems consistent with the essentially continuous montane-forest belt of the Albertine Rift, which can support a single, wide-ranging species (Demos, 2014). The pronounced mitochondrial divergence between them may reflect ‘ecotypic islands’ within the species, generated by adaptive ecological barriers despite ongoing nuclear gene flow (Cuypers et al., 2022; Thomas et al., 2025). However, considering mito-nuclear discordance where mitochondrial markers like *CYTB* may indicate deep lineage splits and limited distributions, while nuclear loci fail to show the same divergence due to incomplete lineage sorting, slow sorting of nuclear loci, or introgression (Toews and Brelsford, 2012; Platt et al., 2017), it is equally plausible that these late-diverging MOTUs are narrowly distributed, recently isolated species rather than widespread montane-forest belts. Unlike the *CYTB* tree, MOTU16 was resolved at UCE as an early-splitting sister to the Ethiopia–Upper Guinea group. In its range, it is sympatric with the widely sampled MOTU24[cf. *microtis*], a pattern typical of the Eastern African montane archipelago, where successive colonization by earlier- versus later-diverging lineages generates layered assemblages, reinforcing their role as climatic refugia and engines of rodent

diversification (Demos et al., 2014; Demos et al., 2015; Couvreur et al., 2021; Huntley, 2023). On the other hand, the MOTU24[cf. *lorraineus*] + MOTU12 group forms a well-supported pair separated by 11.7% *CYTB* K2P distance, well above typical intraspecific divergences, and similarly strongly supported as divergent in the UCE tree. The contiguous east–west distribution abutting the Sudanian savanna belt compares to documented deep population splits within single species in other murid rodents such as *Mastomys*, *Grammomys*, *Arvicanthis*, *Lemniscomys*, and *Gerbilliscus* (Brouat et al., 2009; Granjon et al., 2012; Bryja et al., 2014c; Bryja et al., 2019; Hánová et al., 2021; Bryja et al., 2024).

In the late-diverging MOTUs of the pan-African lineage, the second group comprises East African samples (MOTU10, MOTU13, MOTU17) + (MOTU21, MOTU22, MOTU23, MOTU24[cf. *microtis*]), which are the latest divergences within the genus. The MOTU10 + MOTU13 + MOTU17 + MOTU18 group is a slightly early split, with each MOTU meeting genetic–species criteria and exhibiting coherent geographic envelopes, yet no type locality obviously links them to names available in the current accepted taxonomy. Several dormant *G. murinus* synonyms that are linked to early classical collections (Mammal Diversity Database, 2025) may apply and could be re-evaluated and resurrected to full species ranks, representatives of these MOTUs. However, together with increasing evidence that several Afrotropical rodents previously thought to be widespread are but complexes of range-restricted lineages (Voelker et al., 2021; Bryja et al., 2024), *G. murinus* may be confined to southern African refugia. The potential absence of the species’ representatives in our dataset likely reflects genuine geographic restriction rather than a sampling artifact. Comprehensive genomic and morphological sampling that extends southward to the Cape Province will be key to determining whether *G. murinus* represents a single southern lineage or a complex of cryptic taxa within the genus.

4.2 *Graphiurus* evolutionary divergence history

We recovered an early Miocene split of the African dormouse from other dormice, establishing an ancient lineage within Gliridae (Montgelard et al., 2003). The emerging picture is that a proliferation of species did not immediately follow this initial split, as the ancestor persisted for a long time with little internal speciation, likely due to an extended period of environment-driven diversification constraints (Moen and Morlon, 2014), albeit it could also be an artefact of the poor fossil record. The great bulk of species-level diversity in *Graphiurus* accumulated during a Pliocene–Pleistocene radiation pulse that overlapped a marked rise in continental climatic variability and stepwise aridification beginning ca. 3.5 Ma and peaking around 2.8, 1.7, and 1.0 Ma (deMenocal, 1995; Fabre et al., 2012; Somelar et al., 2018; Cohen et al., 2022). It seems that Pliocene–Pleistocene landscape changes, rather than their deep-time origin, were stronger radiation drivers for the African dormice (Ramírez-Pedraza et al., 2024). As climatic oscillations repeatedly fragmented tropical forest habitats in which dormice first evolved (Montgelard et al., 2003), *Graphiurus* seems to have allopatrically speciated within forest refugia and progressively expanded into savanna–woodland mosaics (Fig. S11) where novel niches emerged (Blumenthal et al., 2017; Lepre and Quinn, 2022). For example, the most nested MOTUs typify mesic savanna–woodland systems, whereas ‘MOTU8’ is confined to the xeric Somali–Masai grasslands, an area

known for arid endemics such as the unstriped ground squirrel *Xerus rutilus* (deMenocal, 2012; Wilson et al., 2017).

The earliest internal divergence within *Graphiurus* (Graphiurinae), separating the West African lineage from the pan-African lineage, coincides with the Middle Miocene Climatic Optimum, a period of globally warmer and wetter climates (Scotese et al., 2021) during which forests expanded across Africa, creating new habitats (Couvreur et al., 2021) that likely provided new ecological opportunities, facilitating the diversification and radiation of rodents such as dormice. The ancestral *Graphiurus*, constrained by relatively harsher conditions, may have radiated during periods of elevated landscape heterogeneity, adapting to broad niche tolerances and colonizing a wide pan-African range early on, without elevated diversification (Reaney et al., 2018).

After the deep West African lineage MOTUs, divergences within the broadly distributed group resumed in the late Miocene following an ~6.8 Ma gap: MOTU4, MOTU6, MOTU5, and MOTU7. The divergence hiatus is likely due to unsampled lineages or to paleoclimatic effects on divergence-dispersal dynamics primarily, although such patterns can also arise from extinction, and population-structuring processes that inflate coalescent times relative to speciation times (Templeton, 1981; Edwards and Beerli, 2000; Mason et al., 2020; Warnock et al., 2020). Africa's late Miocene experienced increasingly open and dry habitats, as grasslands spread and continuous forests contracted, conditions that have been linked to diversification pulses in many African taxa (Faith et al., 2024), including rodents (Mikula et al., 2021). For *Graphiurus*, late Miocene conditions seem to have prompted vicariant speciation in a stepwise fashion, as each of the deep, earliest-diverging MOTUs suggests that lineages retreated into or survived in different isolated habitat refugia (Fig. S11) following the late Miocene aridification and C4-savanna expansion, whose ecological impacts were intensified by uplifts in the East African region (Montgelard et al., 2003; Sepulchre et al., 2006; Hopley et al., 2019). Notably, the branching pattern, where ancestral MOTUs repeatedly split single descendant lineages rather than synchronously radiating, suggests sequential vicariance and limited early adaptive radiation, as expected under progressive habitat fragmentation of montane/forest through the Miocene–Pliocene; comparable stepwise or refugial splits are documented in other Afrotropical MOTUs and small mammals (Bryja et al., 2014a; Aduse-Poku et al., 2021; Couvreur et al., 2021; Dagallier et al., 2024; Dietz et al., 2024).

While the common ancestor contributing to most of the extant dormice's current diversity dates to the Plio-Pleistocene, a significant shift in the evolutionary tempo of *Graphiurus* occurred at the cusp of the Pliocene. Over this period, as some diverging lineages remained restricted (such as MOTU8 and MOTU21), others went on to colonize broad habitats across west-central Africa to East Africa (such as MOTU22, MOTU23, and MOTU24[cf. *microtis*]). The relatively warm, oscillating climate of the Pliocene led to both expansion and contraction of forests, periodically increasing connectivity among forest habitats and promoting the emergence of montane forest 'islands' (deMenocal, 2004; Salzmann et al., 2009). *Graphiurus* likely utilized these dynamics through adaptive radiation, with some lineages moving back into the East African highlands, others spreading along the forest-savanna mosaic of the Sudanian zone, and others reaching southern Africa. Moreover, the internal divergences within the two main clusters of *Graphiurus* emerging

by the end of the Pliocene (the group stretching from Ethiopia to Upper Guinea, and the Eastern to Southern Africa group centered in the East African highlands) continued into the Pleistocene, which is coincident with repeated expansion and contraction of habitable zones due to climate fluctuations, as observed in other small mammal species across the continent (Bryja et al., 2014a; Bryja et al., 2014b; Jacquet et al., 2015; Mizerovská et al., 2019).

4.3 *Graphiurus* ancestral biogeographic range dynamics

Our evolutionary divergence and biogeographic history reconstructions suggest certain refugial regions influenced the divergence and spread of *Graphiurus* across sub-Saharan Africa. The Upper Guinean rainforest likely served as a cradle for *Graphiurus* before it expanded throughout the Afrotropics, aligning with its established role as a Miocene–Pliocene refugium for rodents and plants (Bostoen et al., 2015; Garcin et al., 2018; Maley et al., 2018; Madikiza et al., 2019; Ernst et al., 2025). From this western ‘center’, the genus appears to have expanded along two main points. The deeply divergent MOTUs of the Eastern Arc Mountains and the volcanic highlands of Kenya and Tanzania indicate long-term persistence in the Eastern Afromontane highlands through arid intervals, followed by localized speciation (Bryja et al., 2014a; Demos et al., 2014). Temporal mid-elevation forest strands may then have enabled northeastward expansions into Ethiopia, producing a mosaic of ancient endemics and younger lineages (Taylor et al., 2011; Onditi et al., 2021). On the other hand, the sister relationship between the Ethiopia-to-Upper Guinea MOTUs suggests that moist interglacials created gallery-forest archipelagos across a Sudan–Guinea savanna corridor, allowing dormice to disperse through forest-edge habitats. Consequent arid phases would then have reimposed the barrier, fostering East–West allopatry (Brouat et al., 2009; Granjon et al., 2012; Bryja et al., 2014c; Bryja et al., 2019; Hánová et al., 2021; Bryja et al., 2024).

Two parsimonious dispersal hypotheses align with the reconstructed ancestral range and phylogenetic patterns. (i) An early west-central expansion into southern Africa was followed by later northeastern range shifts toward the East African mountains during periods of increased habitat connectivity (Stanley et al., 1998; Bostoen et al., 2015) as noted in other low-dispersal African small mammals, such as the laminated-toothed rats (Otomyini tribe). Several lineages of the Otomyini tribe originated in southern Africa, from where they experienced multiple northward colonization waves associated with Plio-Pleistocene climatic oscillations (Montgelard et al., 2003; Taylor et al., 2004; Taylor et al., 2011). (ii) Conversely, early colonization of the Ethiopian Highlands/East African Rift montane forests was followed by southward expansion into southern Africa (Jacquet et al., 2015; Onditi et al., 2021; Dianat et al., 2023; Dianat et al., 2024). The late Miocene age of the earliest-branching southern lineage (MOTU4) aligns with repeated Neogene–Quaternary fluctuations in African hydroclimate and forest cover that alternately promoted connectivity and isolation. This scenario matches the recent palaeoclimatic niche modeling in mouse shrews genus *Myosorex*, which supports a southward stepping-stone history with repeated glacial-range expansions and interglacial fragmentation, and suggests a route that allowed the genus to island-hop southward from Central and East Africa along the Rift (Willows-Munro and Matthee, 2009; Pahad et al., 2023). Regions like the Livingstone Mountains act as potential stepping stones between major montane blocks, such as the Eastern Arc Mountains and the Southern Rift Mountains, while gaps like the Makambako Gap and Rukwa rift serve as

intermittently permeable barriers (Cuypers et al., 2022), providing a corridor-and-filter framework for either of the directional hypotheses. Similarly, speciation within the widespread shrews of the *Crocidura olivieri* complex was driven by Pleistocene climate changes through forest refuges, river barriers, and environmental differences that caused deep geographic divisions and range expansions (Dianat et al., 2023; Dianat et al., 2024).

Between these refugial regions, the Congo Basin appears to have facilitated peripheral dispersal of *Graphiurus* lineages, with the northern Sudanian frontier and the southern miombo woodlands occasionally forming semi-open corridors that allowed eastward and southwestward movements (Nicolas et al., 2011; Bohoussou et al., 2015; Hassanin et al., 2015; Jacquet et al., 2015; Stribna et al., 2019; Onditi et al., 2021). This argument aligns with our ancestral-range reconstructions, which consistently suggest that *Graphiurus* range evolution was driven mainly by founder-event speciation during cladogenesis. It also supports the idea that the lack of phylogeographic intermediates between the deep-diverging West African lineages and the younger southeastern African MOTUs may be due to extinction, incomplete sampling, or both. In any case, more extensive field surveys with targeted sampling are essential to fill this gap.

4.4 Caveats, limitations, and interpretive uncertainty

Several methodological and sampling-related considerations qualify our interpretation of phylogenetic concordance, divergence-time estimates, and historical biogeographic reconstructions in *Graphiurus*.

- (i) The strong topological concordance between the *CYTB* tree and the UCE phylogeny is consistent with a shared underlying species-tree signal and suggests that pervasive species-level incomplete lineage sorting (ILS) at the nodes currently resolved is unlikely. However, some ILS is still expected under the multispecies coalescent, especially across short internodes, and cannot be ruled out without explicit gene-tree/species-tree discordance analyses (Degnan and Rosenberg, 2009). Where mitochondrial–nuclear discordance is detected, it should be treated as a biologically interpretable signal; such as mito-nuclear incongruence which commonly arises from recent introgression/hybridization, ILS, and demographic processes, including sex-biased dispersal, and need not imply analytical artifacts by default (Toews and Brelsford, 2012).
- (ii) Divergence-time estimates, while internally precise, remain conditional on calibration and clock-model assumptions; calibration uncertainty in particular can shift absolute dates, so interpretations should emphasize relative ordering and overlap of posterior credible intervals rather than single-point ages (Ho and Phillips, 2009).
- (iii) Our historical biogeographic model comparison favored founder-event speciation, but we acknowledge the ongoing conceptual and statistical debate over the interpretation of the *J* parameter and over model-selection comparisons (Matzke, 2014; Matzke, 2022). Accordingly, we treat founder-event inferences as plausible hypotheses rather than definitive process claims (Matzke, 2014). However, despite remaining sampling gaps, particularly in southern Africa, the reconstructed history of *Graphiurus* is

currently compatible with either a rare founder-event (jump) dispersal or a stepwise (stepping-stone) range expansion across adjacent regions.

4.5 Conclusion

We provide the first comprehensive evolutionary framework for the African dormice (genus *Graphiurus*), revealing an ancient lineage that underwent continued radiation. The monophyly of *Graphiurus* and its early divergence within Gliridae are supported. However, genetic diversity within the genus does not align with current taxonomy, indicating substantial unrecognized evolutionary distinctiveness and necessitating an integrative taxonomic revision. The evolutionary history of *Graphiurus* reflects alternating phases of paleoclimatic stability and instability, where long-term refugia preserved deep Miocene lineages, whereas later climatic volatility promoted pulses of dispersal and speciation. In this respect, *Graphiurus* mirrors a broader pattern of extended stasis punctuated by episodic diversification across multiple climate regimes seen in other African fauna, such as some groups in the shrew genus *Crocidura* (Dianat et al., 2023). Across MOTUs, we found distribution discontinuities that conflict with several traditionally widespread species (e.g., *G. kelleni*, *G. microtis*, and *G. murinus*). The only broadly distributed complex we recover is the MOTU22–MOTU23–MOTU24[cf. *microtis*] cluster, consistent with a single-species complex. Moreover, whether West African MOTUs that extend into the Congolian belt reflect ancient dispersals or more recent events remains unresolved. Denser geographic sampling, especially in poorly surveyed Afrotropical regions for this genus, will be critical for testing cryptic diversity, clarifying lineage gaps (extinction vs. undersampling), and refining biogeographic hypotheses. Our findings establish a robust baseline for future revision of the *Graphiurus* taxonomy, refine our understanding of ecological differentiation among MOTUs, and test lineage-specific biogeographical hypotheses.

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Data statement

Consistent with the journal's data availability guidelines, the new cytochrome b gene sequences have been deposited in GenBank (accession numbers: PX513364–PX513453). The raw data, analysis scripts, and UCE sample-specific contigs, alignments, and trees are available in Zenodo (DOI: 10.5281/zenodo.17336599) and GitHub (<https://github.com/okennott/African-dormice>).

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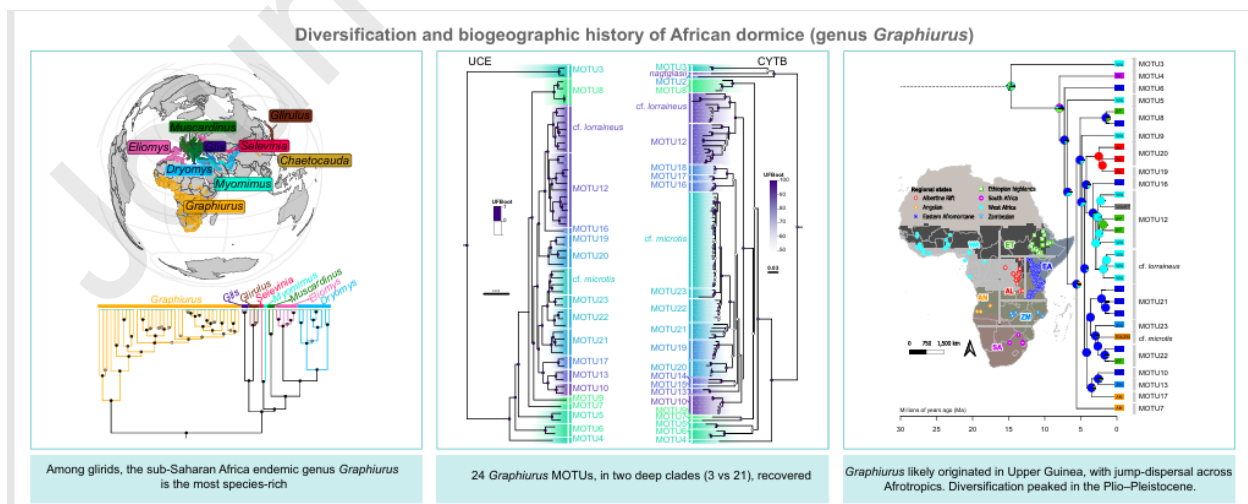
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Author contributions

KOO: conceptualization; methodology; investigation; formal analysis; data curation; resources; writing – original draft; writing – review & editing. TCD: conceptualization; methodology; resources; writing – review & editing. JCK-P: methodology; investigation; resources; writing – review & editing. VN: methodology; investigation; resources; writing – review & editing. OM: conceptualization; methodology; investigation; resources; writing – review & editing. AB: methodology; resources; writing – review & editing. AH: resources; writing – review & editing. AM: resources; writing – review & editing. JB: conceptualization; methodology; data curation; resources; writing – review & editing; funding acquisition.

Highlights—

- The first comprehensive phylogenomic dataset clarifies evolutionary relationships and highlights diversity within *Graphiurus*.
- 24 highly supported genetic lineages in two deeply divergent, reciprocally monophyletic clades at the level of separate genera identified.
- Divergence started in the early Miocene, but major diversification events aligned temporally with Pliocene–Pleistocene climatic fluctuations.
- Long-term habitat refugia in the Eastern African highlands were critical for lineage persistence.
- An integrative taxonomic revision of the genus is needed.



Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: